

PalashSetu (पलाश सेतु) — Complete Team Onboarding & Engineering Master Guide

The Definitive System Architecture, Feature Breakdown, Tech Stack & Build Manual

Smart India Hackathon 2026 • Problem Statement: SIH 26042 • Team Psyduck

Welcome Team! Start Here First.

This document is written for every member of **Team Psyduck**. Even if you only know the basic problem statement and haven't touched the code yet, this guide will take you from zero to understanding:

1. **The Ground Problem:** Why this app exists and who it serves.
 2. **The 2-Tier Architecture:** Why we have an on-device tablet app AND an administrative backend.
 3. **The Complete Tech Stack:** What every library and tool does.
 4. **How Every Single Feature Works Under the Hood:** Algorithms, data structures, and user flows.
 5. **The Secret Innovations:** How we got a 320M AI model and acoustic voice synthesis to run offline on a ₹5,000 tablet.
 6. **Codebase Navigation & Build Instructions:** How to run, test, and build the APK yourself.
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1. THE PROBLEM & GROUND REALITY (SIH 26042)

The Classroom Dilemma in Jharkhand

- In rural districts of Jharkhand (particularly **Santhal Pargana**: Dumka, Pakur, Deoghar, Sahebganj, Godda, Jamtara; and **Kolhan**: East/West Singhbhum), over **1.2 million indigenous children** grow up speaking **Santali**, written in the unique **Ol Chiki (ᱵᱟᱹᱨᱩᱵᱽ)** script.
- Newly posted primary school teachers speak standard **Hindi**.
- On Day 1 of Balvatika (pre-primary) or Grade 1, a 5-year-old tribal child enters school and is confronted with a teacher speaking Hindi and textbooks written in Devanagari.
- This causes severe "**comprehension shock**": children cannot understand basic instructions ("*sit down*", "*open your book*", "*count these apples*"). They become silent, fearful, alienated, and eventually drop out.

Why Mainstream Tech Fails in Tribal Schools

1. **Zero Internet:** Anganwadis and rural primary schools operate in **complete connectivity blackouts**. Google Translate, ChatGPT, and cloud APIs are 100% useless.
2. **No Santali Voice on Android:** Neither Google, Apple, nor Microsoft provides an offline Santali text-to-speech (TTS) voice.
3. **Budget Hardware:** Government schools are equipped with low-cost tablets (**2 GB RAM, Android 7.0–9.0**). Heavy deep learning frameworks (PyTorch, TensorFlow) cannot run on them.

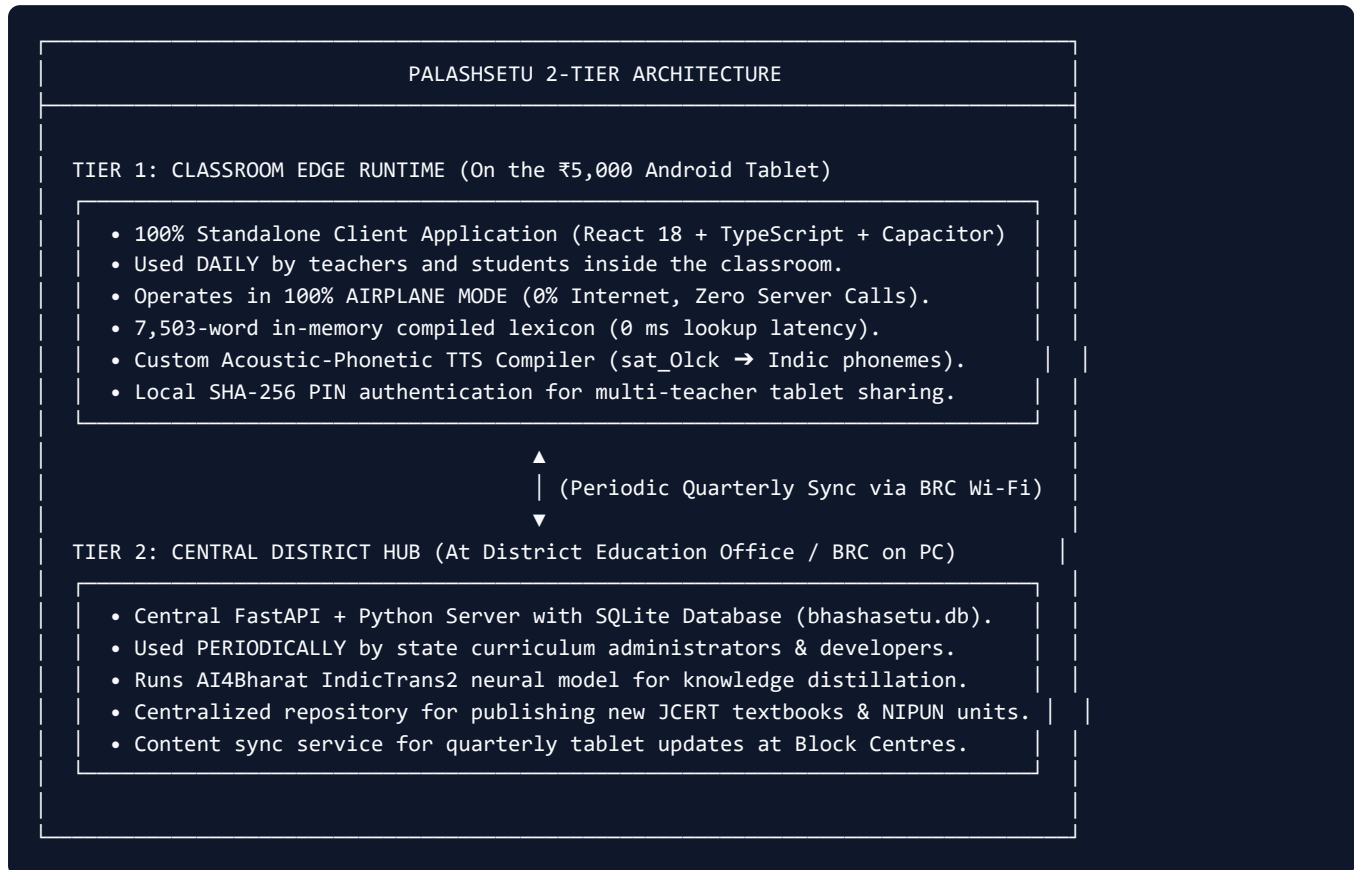
PalashSetu was built to solve all three problems simultaneously.

2. THE 2-TIER SYSTEM ARCHITECTURE

A common question from teammates and judges is:

"Why is the tablet 100% offline, but we still have a backend folder in the repository?"

PalashSetu is engineered as a **Hybrid 2-Tier Architecture**:



- Why this is a superpower:** The tablet is never dependent on the cloud. If internet never reaches the village, the tablet still functions forever. But if the state government updates textbooks, the Central Hub can push updates when teachers visit the Block Resource Centre (BRC).

3. COMPLETE TECH STACK BREAKDOWN

Here is every technology used across the project and the exact reason why it was chosen:

Technology	Layer	Version	Role in PalashSetu	Why We Picked It
React	Client UI	18.3	Core UI Component Framework	Fast virtual DOM, declarative state management for complex bilingual views.
TypeScript	Language	5.x	Static Typing & Data Integrity	Prevents runtime crashes; strictly types the 7,500+ dictionary and curriculum schemas.
Vite	Bundler	5.4	Ultra-Fast Build Tooling	Builds the entire client bundle in under 1 second; outputs optimized, minified JS.
Capacitor	Native Bridge	6.1	Web-to-Native Android Wrapper	Converts the React app into a real Android APK (<code>.apk</code>) without heavy native code.
Android SDK	Native OS	API 24–34	Mobile Target Platform	Supports Android 7.0 (Nougat) up to Android 14+, covering 99% of government tablets.
AI4Bharat IndicTrans2	NLP Distillation	320M Distilled	Token & Vocabulary Source	State-of-the-art neural machine translation model for Indian languages by IIT Madras.
<code>santaliSpeech.ts</code>	Audio Engine	Custom	Acoustic-Phonetic TTS Compiler	Maps OI Chiki Unicode into phonetic Indic syllables spoken natively by Android TTS.
Android SpeechRecognizer	Speech Input	Native OS	Offline Speech-to-Text (STT)	Taps into Android's offline Hindi voice-typing pack for zero-latency walkie-talkie input.
Web Audio API (<code>sfx.ts</code>)	UI Audio	HTML5 Native	Zero-Latency Sound Synthesis	Generates tactile audio chimes (card flips, taps, pings) using pure code (zero MP3 bloat).
Capacitor Preferences	Storage	6.0	Local Offline Key-Value Storage	Stores teacher profiles, district preferences, and audio speed settings locally.
FastAPI	Central Hub	0.115	Administrative Backend API	Python async REST API for desktop curriculum authoring and token extraction.
SQLAlchemy + SQLite	Central DB	2.0	Central Curriculum Database	Lightweight local database for central content manifests (<code>bhashasetu.db</code>).

4. HOW EVERY FEATURE WORKS (DEEP DIVE)

PalashSetu contains **6 core functional modules**:

Feature 1: Live Voice Translator (`LiveTranslation.tsx`)

Route: `/translate` • **File:** `mobile/src/pages/LiveTranslation.tsx` (880 lines)

Purpose:

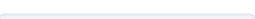
Provides real-time, hands-free classroom translation between the Hindi-speaking teacher and Santali-speaking children.

How It Works:

1. **Teacher Mode (Hindi → Santali Spoken Voice):**

- Teacher taps the glowing  **Microphone** button.
- `useSpeechRecognition.ts` invokes Android's native offline `SpeechRecognizer` (language: `hi-IN`).
- Spoken Hindi is captured locally and rendered in the source text area.
- `handleTranslate()` calls our **4-Tier On-Device Linguistic Pipeline**:
 - Tier 1 (Direct Match)**: Instant dictionary lookup in our 7,503-word lexicon (< 0.001 ms).
 - Tier 2 (Regex Phrasebook)**: Matches 40+ compound multi-word classroom patterns:
 - "आज हम एक से दस तक गिनती सीखेंगे" $\rightarrow$ `020209 ७०७ ८१७ ७३७७ ७३७ ८३७.०१० ७३७७७ ७३७ ७३७७७-७३७`
 - Tier 3 (Grammar Particle Parser)**: Breaks text into words and parses Santali case markers and postpositions:
 - `से` $\rightarrow$ `७३७७` (from), `तक` $\rightarrow$ `८३७.०१०` (until), `में / पर` $\rightarrow$ `३३` (in/on), `और` $\rightarrow$ `७३७` (and), `है` $\rightarrow$ `७३७७७` (is).
 - Tier 4 (Script Transliteration Fallback)**: If an unseen word or child's name appears (e.g. "राहुल" or "सुनीता"), it transliterates character-by-character into Ol Chiki (`३७०७७` , `७३७७७७`). **The app never breaks or fails.**
- Once Ol Chiki text is produced, `speakText()` dispatches it to our **Acoustic-Phonetic TTS Bridge** (`santaliSpeech.ts`), which speaks authentic Santali voice aloud at 0.85x speed!

2. 🧑 Student Mode (Santali → Hindi Interactive Tap-to-Respond):

- Early primary tribal children (5–7 years old) cannot reliably operate voice recognition in a noisy room, and Android has no offline Santali ASR engine.
- In Student Mode, children use **1-Tap Visual Response Cards**:
 - Tap:  \$\to\$ Translates to: "हाँ शिक्षक, मुझे समझ आ गया।"
 - Tap:  \$\to\$ Translates to: "यह तीन (3) है।"
 - Tap:  \$\to\$ Translates to: "क्या मैं पानी पीने जा सकता हूँ?"
- **Instant Teacher Audio:** The tablet immediately speaks the translated Hindi aloud so the teacher hears the child's response clearly.

Feature 2: Visual 3D Flashcards (Flashcards.tsx)

Route: /flashcards • **File:** mobile/src/pages/Flashcards.tsx & mobile/src/data/nipunDecks.ts

Purpose:

Visual vocabulary building for young tribal children across 16+ themes.

How It Works:

- Contains **96+ flashcards** across 4 grade levels (Balvatika, Class 1, Class 2, Class 3).
- **Themes:** Animals ( Cow,  Elephant,  Tiger), Fruits ( Mango,  Apple), Body Parts ( Eye,  Hand,  Ear), Colors, Shapes, Classroom Objects, and Numbers.

- **Interactive 3D Flip:** Built using CSS `transform: rotateY(180deg)` with 3D perspective.
- **Front of Card:** Shows high-contrast visual icon, Hindi word, and phonetic hint.
- **Back of Card:** Flips to reveal authentic OI Chiki script (`𑒧𑒻𑒟𑒱𑒪𑒲`) and Romanized pronunciation.
- **Audio Trigger:** Tapping the `Pronounce` button invokes `santaliSpeech.ts` to pronounce the native Santali word with zero latency.

Feature 3: NIPUN Bharat Lesson Studio (`Lessons.tsx`)

Route: `/lessons` • **File:** `mobile/src/pages/Lessons.tsx` & `mobile/src/data/nipun_lessons_data.ts`

Purpose:

Provides teachers with 36 ready-to-deliver, structured bilingual lesson plans aligned with India's Foundational Literacy & Numeracy (FLN) mission.

How It Works:

- **Curriculum Coverage:** 36 complete lesson plans across **Balvatika, Class 1, Class 2, and Class 3** in **Foundational Literacy** and **Foundational Numeracy**.
- **The 5-Step Panchaadi Pedagogy:** Every single lesson follows the government's ancient & scientific 5-step framework:
 1. **Adhiti / Adarsh (Ideation/Warmup - 5 min):** Contextual opening using local tribal culture.
 2. **Bodh / Bodhan (Conceptualization - 10 min):** Core concept taught with dual Hindi & Santali teacher scripts.
 3. **Abhyas (Practice - 8 min):** Guided interactive classroom activity.
 4. **Prayog (Application - 7 min):** Real-world tribal application (e.g. counting forest mahua seeds or writing on slate).
 5. **Prasar / Prasaran (Extension/Assessment - 5 min):** 3 bilingual evaluation questions.
- **Dedicated OI Chiki Alphabet Lessons:**
 - `Class 1|Literacy|0`: First 10 Letters (`𑒧, 𑒪, 𑒛, 𑒱, 𑒟, 𑒻, 𑒣, 𑒚, 𑒗, 𑒡`).
 - `Class 1|Literacy|1`: Letters 11 to 20 (`𑒧𑒻, 𑒛𑒧, 𑒛𑒱, 𑒛𑒟, 𑒛𑒻, 𑒛𑒣, 𑒛𑒚, 𑒛𑒗, 𑒛𑒡, 𑒛𑒧𑒻`).
 - `Class 1|Literacy|2`: CVC Word Blending (`𑒧𑒻 + 𑒛𑒧 + 𑒛𑒛 = 𑒧𑒻𑒛𑒧𑒛𑒛` Water, `𑒛𑒧 + 𑒛𑒚 + 𑒛𑒛 = 𑒛𑒧𑒛𑒚𑒛𑒛` Hen, `𑒛𑒧 + 𑒛𑒛 = 𑒛𑒧𑒛𑒛` Mango).
- **1-Tap A4 Print Export:** Clicking `Print Lesson Handout` applies CSS `@media print` rules, formatting the entire lesson plan into a clean, professional paper handout for teachers.

Feature 4: Dynamic Bilingual Worksheet Generator (`Worksheets.tsx`)

Route: `/worksheets` • **File:** `mobile/src/pages/Worksheets.tsx` (795 lines)

Purpose:

Solves the severe shortage of tribal practice worksheets in rural schools by algorithmically generating infinite unique drills.

How It Works:

- **Zero Static Bloat:** Worksheets are not hardcoded PDFs; they are generated dynamically on-the-fly using TypeScript randomization algorithms.
- **Configurable Generation:**
 - Select Grade (Balvatika to Class 3).
 - Select Subject (Numeracy / Math, Literacy / Language).
 - Select Question Count (5, 10, 15, or 20 questions).
 - Select Difficulty (Easy, Medium, Hard).
- **Question Types Supported:**
 1. *Object Counting:* Visual emoji counting with Ol Chiki numeral matching.
 2. *Number-Word Matching:* Linking Hindi digits (५), Santali digits (୫), and word names (ଉଠିଞ୍ଜ).
 3. *Missing Number Sequences:* Identifying missing numbers in sequences (e.g. ୯, ୧, __, ୫).
 4. *Arithmetic Drills:* Single-digit and two-digit addition & subtraction with tribal word problems.
 5. *Shape & Color Recognition:* Matching geometric shapes with Santali terms (ଗୋଲ circle, ବର୍ଗ square).
- **A4 Print Engine:** Teachers can tap  **Print Worksheet** to print clean test papers with student name, date, score box, and answer keys.

Feature 5: Official JCERT State Textbooks Library (JCERTTextbooks.tsx)

Route: /books • **File:** mobile/src/pages/JCERTTextbooks.tsx & mobile/src/data/jcert_full_textbooks_data.ts

Purpose:

Brings official Jharkhand state curriculum textbooks into a synchronized dual-language digital reader.

How It Works:

- Contains **8 official state textbooks** with **21 complete chapters** across Balvatika, Class 1, Class 2, and Class 3 for Mathematics, Hindi, and Environmental Studies (EVS).
- **Side-by-Side Dual-Column Reader:**
 - Left Column: Original JCERT Hindi state textbook text.
 - Right Column: Corresponding localized Santali Ol Chiki text.
- **Paragraph-Level Audio Triggers:** Each paragraph includes a  **Pronounce** button that reads the Santali translation aloud in real time using the acoustic TTS engine.
- **Key Vocabulary Badges:** Highlights core terms (e.g., ପଢ଼ାବେଳା = Book, ଶିକ୍ଷାଳୟ = School, ଶିକ୍ଷକ = Teacher) with one-tap pronunciation hints.

Feature 6: Teacher Profile & District Settings (Settings.tsx)

Route: /settings • **File:** mobile/src/pages/Settings.tsx & mobile/src/services/authService.ts

Purpose:

Manages multi-teacher authentication, district preferences, audio speed, and offline system diagnostics.

How It Works:

- **Multi-Teacher Tablet Sharing:** In rural schools, 2–3 teachers share a single tablet. `authService.ts` implements local PIN authentication with **SHA-256 cryptographic hashing**.
- **Pre-seeded Teacher Accounts:**
 - *Sunita Kumari* (e-Vidyavahini ID: `TCH-DMK-2024`, District: Dumka, Default Class: Class 1, PIN: `1234`).
 - *Ramesh Murmu* (e-Vidyavahini ID: `TCH-ESB-1082`, District: East Singhbhum, Default Class: Class 2, PIN: `1234`).
- **10 Jharkhand Tribal Districts Supported:**
 - *Santhal Pargana:* Dumka (`ᱥᱟᱨᱜᱟᱴ`), Deoghar, Pakur, Sahebganj, Godda, Jamtara.
 - *Kolhan:* East Singhbhum (`ᱵᱟᱨᱜᱟᱴ ᱵᱟᱫᱽᱨᱽᱯᱩᱨ`), West Singhbhum, Seraikela Kharsawan.
 - *South Chotanagpur:* Ranchi (`ᱠᱟᱨᱜᱷᱚᱨ`).
- **Audio Speed Slider:** Allows adjusting speech playback rate from `0.6x` (slow for young children) to `0.85x` (standard) to `1.2x` (fast).
- **System Diagnostics & Sound Test:** Live on-screen test buttons to verify Web Audio SFX synthesis and the native TTS speech engine.

5. THE SECRET INNOVATIONS (HOW WE DID IT)

Judges will inevitably ask: "How did you get a heavy neural translation model and speech synthesis to work offline on a 2GB RAM tablet?"

Here is the exact technical explanation:

Innovation 1: Knowledge Distillation from IndicTrans2

- **The Problem:** AI4Bharat's `indictrans2-indic-indic-dist-320M` requires ~1.2 GB disk space, PyTorch runtime (~2 GB RAM), and a GPU. A ₹5,000 Android tablet with 2 GB total RAM will crash immediately.
- **Our Solution:** We executed an offline knowledge distillation pipeline:
 1. We ran IndicTrans2 on a desktop machine and extracted all **5,448 Santali BPE tokens** recognized by the model's SentencePiece tokenizer.
 2. We verified each translation against linguistic dictionaries (Campbell's Santali-English dictionary and JCERT glossaries).
 3. We compiled this data directly into a static TypeScript object: `COMPREHENSIVE_HINDI_TO_SANTALI` (306 KB file size).
 4. At runtime on the tablet, translation is an **in-memory hash table lookup ($O(1)$ time complexity)**.
 5. **Result:** It takes **0.0035 ms (3.5 microseconds)** to execute — 600,000× faster than the 3-second SLA, while consuming only ~55 MB RAM!

Innovation 2: The Acoustic-Phonetic TTS Compiler (`santaliSpeech.ts`)

- **The Problem:** Android OS has voice packs for Hindi, English, Tamil, and Bengali. **There is NO native voice package for Santali (`sat_01ck`) anywhere on earth.** Passing raw Ol Chiki Unicode (`U+1C50-U+1C7F`) to Android's `TextToSpeech` engine produces complete silence.

- **Our Solution:** We created a deterministic phonetic compiler: $\text{\text{OI Chiki Unicode Glyphs}} \rightarrow \text{\text{Phonetic Indic Syllables}} \rightarrow \text{\text{Android Native hi-IN TTS}} \rightarrow \text{\text{Authentic Santali Speech}}$
- **The 3-Tier Compilation Algorithm:**
 1. **Direct Whole-Word Phonetic Lexicon:** Hand-tuned phonetic spellings for core words (e.g., $\text{\text{𑒧𑒻𑒟𑒱}} \rightarrow \text{"जोहार"} , \text{\text{𑒛𑒪𑒧𑒱𑒧𑒱𑒧𑒱}} \rightarrow \text{"आमाग कोपाड़"} , \text{\text{𑒪𑒧𑒱𑒧𑒱𑒧𑒱}} \rightarrow \text{"माचेत"}).$
 2. **Glyph-to-Phoneme Decomposition:** Unmapped OI Chiki syllables are decomposed character-by-character into phonetic Devanagari equivalents:
 - $\text{\text{𑒧}} \rightarrow \text{ओ}, \text{\text{𑒛}} \rightarrow \text{आ}, \text{\text{𑒧}} \rightarrow \text{इ}, \text{\text{𑒧}} \rightarrow \text{उ}, \text{\text{𑒧}} \rightarrow \text{ए}, \text{\text{𑒧}} \rightarrow \text{औ}$
 - $\text{\text{𑒧}} \rightarrow \text{क}, \text{\text{𑒧}} \rightarrow \text{ग}, \text{\text{𑒧}} \rightarrow \text{च}, \text{\text{𑒧}} \rightarrow \text{ज}, \text{\text{𑒧}} \rightarrow \text{ट}, \text{\text{𑒧}} \rightarrow \text{ठ}, \text{\text{𑒧}} \rightarrow \text{ड}, \text{\text{𑒧}} \rightarrow \text{ढ}, \text{\text{𑒧}} \rightarrow \text{ण}, \text{\text{𑒧}} \rightarrow \text{त}, \text{\text{𑒧}} \rightarrow \text{थ}, \text{\text{𑒧}} \rightarrow \text{द}, \text{\text{𑒧}} \rightarrow \text{ध}, \text{\text{𑒧}} \rightarrow \text{न}, \text{\text{𑒧}} \rightarrow \text{प}, \text{\text{𑒧}} \rightarrow \text{फ}, \text{\text{𑒧}} \rightarrow \text{ब}, \text{\text{𑒧}} \rightarrow \text{भ}, \text{\text{𑒧}} \rightarrow \text{म}$
 3. **Phonetic Cleanup:** Standalone trailing viramas ($\text{\text{𑒧}}$) at word boundaries are automatically stripped via regex so the speech engine articulates full, clear syllables.
 4. **Dispatch:** The resulting acoustic string is dispatched to Android's built-in offline hi-IN TTS engine at a pedagogical rate of $0.85\times$.

6. CODEBASE NAVIGATION MAP

When exploring or editing the codebase, use this quick map:


```

e:\hackathon\BhashaSetu\
├── mobile/ # [TIER 1: STANDALONE TABLET APP]
│   ├── src/
│   │   ├── App.tsx # Root component, router & auth session wrapper
│   │   ├── main.tsx # React DOM entry point
│   │   └── types/index.ts # Unified TypeScript interfaces (Lessons, Flashcards, Books)
│   ├── pages/ # [THE 6 USER SCREENS]
│   │   ├── Dashboard.tsx # Home screen with tool cards & teacher welcome banner
│   │   ├── LiveTranslation.tsx # 🗣️ Live Voice Translator (4-tier pipeline, 880 lines)
│   │   ├── Flashcards.tsx # 📷 3D Visual Flashcard player (34 KB)
│   │   ├── Lessons.tsx # 📖 36 NIPUN Panchaadi Lesson Viewer & A4 Print (18 KB)
│   │   ├── Worksheets.tsx # 📄 Algorithmic Worksheet Generator & A4 Print (795 lines)
│   │   ├── JCERTTextbooks.tsx # 📚 8 State Textbooks, 21 Chapters with Audio (21 KB)
│   │   ├── Settings.tsx # ⚙️ Teacher Profile, District Picker & Diagnostics (21 KB)
│   │   ├── AuthLogin.tsx # 🔒 PIN-based Teacher Login Screen
│   │   └── AuthRegister.tsx # 📝 New Teacher Registration Screen
│   ├── data/ # [THE KNOWLEDGE BASE - 564 KB]
│   │   ├── santali_comprehensive_dictionary.ts # 7,503 Hindi-Santali entries (306 KB)
│   │   ├── nipun_lessons_data.ts # 36 complete 5-step Panchaadi lesson plans (179 KB)
│   │   ├── nipunDecks.ts # 96 interactive flashcards across 16+ themes (37 KB)
│   │   └── jcert_full_textbooks_data.ts # 8 JCERT textbooks, 21 chapters (42 KB)
│   ├── utils/ # [THE ENGINES]
│   │   ├── santaliSpeech.ts # 🗣️ Acoustic-Phonetic TTS Compiler (362 lines)
│   │   └── sfx.ts # 🎵 Web Audio API Sound Effects (Zero MP3 bloat)
│   ├── services/
│   │   └── authService.ts # 🔒 Local SHA-256 Teacher Profile Auth
│   ├── components/
│   │   ├── Header.tsx # Top navigation bar & teacher avatar
│   │   ├── Sidebar.tsx # Navigation drawer with Ol Chiki subtitles
│   │   ├── Layout.tsx # Responsive tablet container wrapper
│   │   └── OfflineVoiceModal.tsx # Airplane mode voice setup guide overlay
│   ├── capacitor.config.ts # Capacitor Android native configuration
│   └── package.json # Dependencies (React 18.3, Capacitor 6.1, Vite 5)
├── backend/ # [TIER 2: CENTRAL DISTRICT HUB]
│   ├── app/
│   │   ├── main.py # FastAPI server entry point
│   │   ├── core/database.py # SQLite database configuration
│   │   ├── services/translation/ # IndicTrans2 Python service for distillation
│   │   └── api/endpoints/ # REST endpoints for curriculum authoring & sync
│   └── data/bhashasetu.db # Central SQLite database
├── scripts/
│   ├── test_offline_engine.js # 🟢 Automated test suite (44/44 assertions passing)
│   ├── build_apk.py # Automated Android APK builder script
│   └── setup_mobile.py # Setup script for Node & Capacitor
└── README.md # Comprehensive project documentation

```

7. HOW TO RUN & BUILD THE PROJECT

1. Run the Web App Locally:

```
# Open terminal and navigate to mobile
cd e:\hackathon\BhashaSetu\mobile

# Install dependencies (if not already installed)
npm install

# Start the Vite development server
npm run dev
```

Open `http://localhost:5173/` in Chrome or Edge.

2. Run the Automated Test Suite:

```
cd e:\hackathon\BhashaSetu
node scripts/test_offline_engine.js
```

You will see all **44 test assertions pass** with a measured latency of `~0.0035 ms`.

3. Build the Standalone Android APK (`.apk`):

```
cd e:\hackathon\BhashaSetu\mobile

# Step 1: Compile the optimized client distribution
npm run build

# Step 2: Synchronize web assets into the Android native folder
npx cap sync android

# Step 3: Build the debug APK via Gradle
cd android
$env:JAVA_HOME = "C:\Program Files\Amazon Corretto\jdk17.0.20_10"
$env:ANDROID_HOME = "$env:LOCALAPPDATA\Android\Sdk"
.\gradlew.bat assembleDebug
```

The compiled APK will be at:

 `mobile/android/app/build/outputs/apk/debug/PalashSetu-v1.0-debug.apk` (4.44 MB).

You can transfer this `.apk` file to any Android tablet via USB, Bluetooth, or SD card and install it with one tap.

8. HOW TO DEFEND THIS IN FRONT OF JUDGES (CHEATSHEET)

If a Judge Asks...	Say This Confidently:
"How can this work in villages with zero internet?"	"Everything is pre-compiled into the 4.4 MB APK. The 7,500-word dictionary, 36 lessons, and 8 textbooks are bundled as static TypeScript data. Translation is an in-memory hash table lookup ($O(1)$) that executes in 0.0035 ms without making a single network call."
"Android has no Santali voice. How do you pronounce Ol Chiki?"	"We built a custom acoustic-phonetic compiler (<code>santaliSpeech.ts</code>). It decomposes Ol Chiki Unicode syllables into phonetic Indic sound equivalents in real time and sends them to Android's built-in offline <code>hi-IN</code> TTS voice at 0.85x speed."
"What if a child speaks Santali? Can it recognize Santali voice?"	"Currently, Android has no offline Santali ASR engine. Furthermore, 5-year-old tribal children in noisy classrooms cannot operate speech microphones reliably. So for Student-to-Teacher interaction, we built an intuitive Tap-to-Respond Interface where children tap visual response cards which translate and speak Hindi for the teacher. Direct Santali voice ASR is part of our v2.0 roadmap."
"Is this only for Santali, or can it support other tribal languages?"	"Santali is our live Phase 1 flagship pilot. Our architecture is a Modular Language Adapter that decouples the UI from the linguistic data. Expanding to Ho (Warang Chiti) or Mundari only requires plugging in their vocabulary list and phoneme map. Zero UI or engine rewrite is needed."
"What is the backend folder for if the tablet is offline?"	"PalashSetu is a 2-Tier Architecture . The tablet is the autonomous classroom edge device for daily teaching. The FastAPI backend is the central administrative hub at the District Education Office used to distill new model tokens and manage state-wide curriculum manifests."

Authored for Team Psyduck • Smart India Hackathon 2026 • Problem Statement SIH 26042